

GLASGOW COLLEGE UESTC

Main Paper - Fall 2022-2023

REAL TIME COMPUTING & ARCH (UESTC4019)

Date: 11th December 2023

Time: 9:30 – 11:30

Attempt all PARTS. Total 100 marks

**Use one answer sheet for each of the questions in this exam.
Show all work on the answer sheet.**

**Make sure that your University of Glasgow and UESTC Student Identification
Numbers are on all answer sheets.**

**An electronic calculator may be used provided that it does not allow text storage
or display, or graphical display.**

**All graphs should be clearly labelled and sufficiently large so that all elements
are easy to read.**

**The numbers in square brackets in the right-hand margin indicate the marks
allotted to the part of the question against which the mark is shown. These
marks are for guidance only.**

Question 1

- (a) (i) Explain what is meant by a real-time system. [2]
- (ii) Define the difference between hard real-time systems and soft real-time and give an example of each type. [4]
- (b) Represent the number $+33.8125_{10}$ in IEEE-754 single precision standard floating point. Show complete work (calculation and steps) to receive full marks. [8]
- (c) Consider the execution of a program that results in the execution of 2 million instructions on a 400-MHz processor. The program consists of four major types of instructions. The instruction mix and the cycles per instruction (CPI) for each instruction type are given in Table Q1c. Calculate the average CPI, and the corresponding MIPS rate. [3]

Instruction Type	Instruction Mix	CPI
Integer Arithmetic	45%	2
Load/store with cache hit	26%	3
Branch	14%	6
Memory reference with cache miss	15%	9

Table Q1(c)

- (d) Now assume that the program can be executed in eight parallel tasks or threads with a roughly equal number of instructions executed in each task. Execution is on an 8-core system, with each core (processor) having the same performance as the single processor originally used. Coordination and synchronization between the parts add an extra 25,000 instruction executions to each task. Assume the same instruction mix as in the example for each task, but increase the CPI for memory reference with cache miss to 12 cycles due to contention for memory.
- (i) Determine the average CPI, and the corresponding MIPS rate. [3]
- (ii) Calculate the speedup factor. [5]

Continued overleaf

Question 2

- (a) Describe the register set of the ARM Cortex M4 processor. [2]
- (b) Explain the rationale behind the existence of multiple buses in the ARM Cortex M series architecture. [4]
- (c) The memory writes timing of a microprocessor specified by its manufacturer is the width of the Write signal that can be determined by $T-50$, where T is the clock period in ns.
 - (i) What width should we expect for the Write signal if the bus clocking rate is 5 MHz? [3]
 - (ii) The data sheet for the microprocessor specifies that the data remain valid for 20 ns after the falling edge of the Write signal. What is the total duration of valid data presentation to memory? [2]
 - (iii) How many wait states should we insert if memory requires valid data presentation for at least 190 ns? [2]
- (d) Consider a system employing interrupt-driven I/O for a particular device that transfers data at an average of 8 KB/s on a continuous basis.
 - (i) Assume that interrupt processing takes about 100 μ s (i.e., the time to jump to the interrupt service routine (ISR), execute it, and return to the main program). Determine what fraction of processor time is consumed by this I/O device if it interrupts for every byte. [4]
 - (ii) Now assume that the device has two 16-byte buffers and interrupts the processor when one of the buffers is full. Naturally, interrupt processing takes longer because the ISR must transfer 16 bytes. While executing the ISR, the processor takes about 8 μ s for the transfer of each byte. Determine what fraction of processor time is consumed by this I/O device in this case. [4]
 - (iii) Now assume that the processor is equipped with a block transfer I/O instruction such as that found on the Z8000. This permits the associated ISR to transfer each byte of a block in only 2 μ s. Determine what fraction of processor time is consumed by this I/O device in this case. [4]

Continued overleaf

Question 3

- (a) Given the quadratic equation $Y = Ax^2 + Bx + C$, write out instructions to compute Y using: [15]

(Hint you can write Ax as $A * x$, and Ax^2 as $A * x * x$ where $*$ implies multiplication)

- (i) One-address instructions,
 - (ii) Two-address instructions,
 - (iii) Three-address instructions,
- (b) Reflect on the responses in Q3a above and discuss the advantages and disadvantages of **zero-address** instructions when compared with **three-address** instructions. [4]

(Hint: discuss in terms of memory and processing time)

- (c) Discuss the differences between Arithmetic instructions, data storage instructions, and control instructions. [6]

Question 4

- (a) List and briefly define three types of superscalar instruction issue policies. [6]
- (b) An application program is executed on a nine-computer cluster. A benchmark program took time T on this cluster. Further, it was found that 25% of T was the time in which the application was running simultaneously on all nine computers. For the remaining time, the application had to run on a single computer.
- (i) Calculate the effective speedup under the aforementioned condition as compared to executing the program on a single computer. Also, calculate the percentage of code that has been parallelized (programmed or compiled so as to use the cluster mode) in the preceding program. [6]
- (ii) Suppose that we are able to effectively use 17 computers rather than 9 computers on the parallelized portion of the code. Calculate the effective speedup that is achieved. [2]
- (c) Consider the problem of scheduling the following sets of tasks (assume that all tasks arrive at time 0).

Task	Period	Execution Time
A	20	5
B	60	10
C	40	10
D	30	5

Table Q4(c)

- (i) From an implementation perspective, what are the advantages/disadvantages of an RMS vs an EDF scheduler? [3]
- (ii) What is the utilization of a single processor running the tasks? [3]
- (iii) Find an RMS schedule for the tasks. [5]

End of question paper